

Lamp Stability Measurements

Setup

To understand how long the monochromator's lamp takes to warm up, and how much its output fluctuates once it has settled, we recorded the lamp spectrum over time in two regimes: a **long-term** run (4 hours, 1 s cadence, 10 scans averaged per sample) capturing warm-up and steady-state drift, and a **fast** run (5 minutes, ~67 Hz, unaveraged) intended to resolve fluctuations on sub-second timescales that the long-term run's averaging would hide. The lamp is a 300 W Xenon source in

a Newport monochromator; spectra were acquired with an Ocean Insight OCEANSR6 spectrometer (s/n SR600410) with the VIS grating positioned at the lamp baseline. All figures and quoted numbers below are reproduced from [lamp_stability.ipynb](#).

Lamp Spectrum

Figure 1 shows the lamp's steady-state spectrum, normalized to its peak intensity, together with the spectrometer's noise floor (mean signal over the first 300 samples of the dark/baseline region).

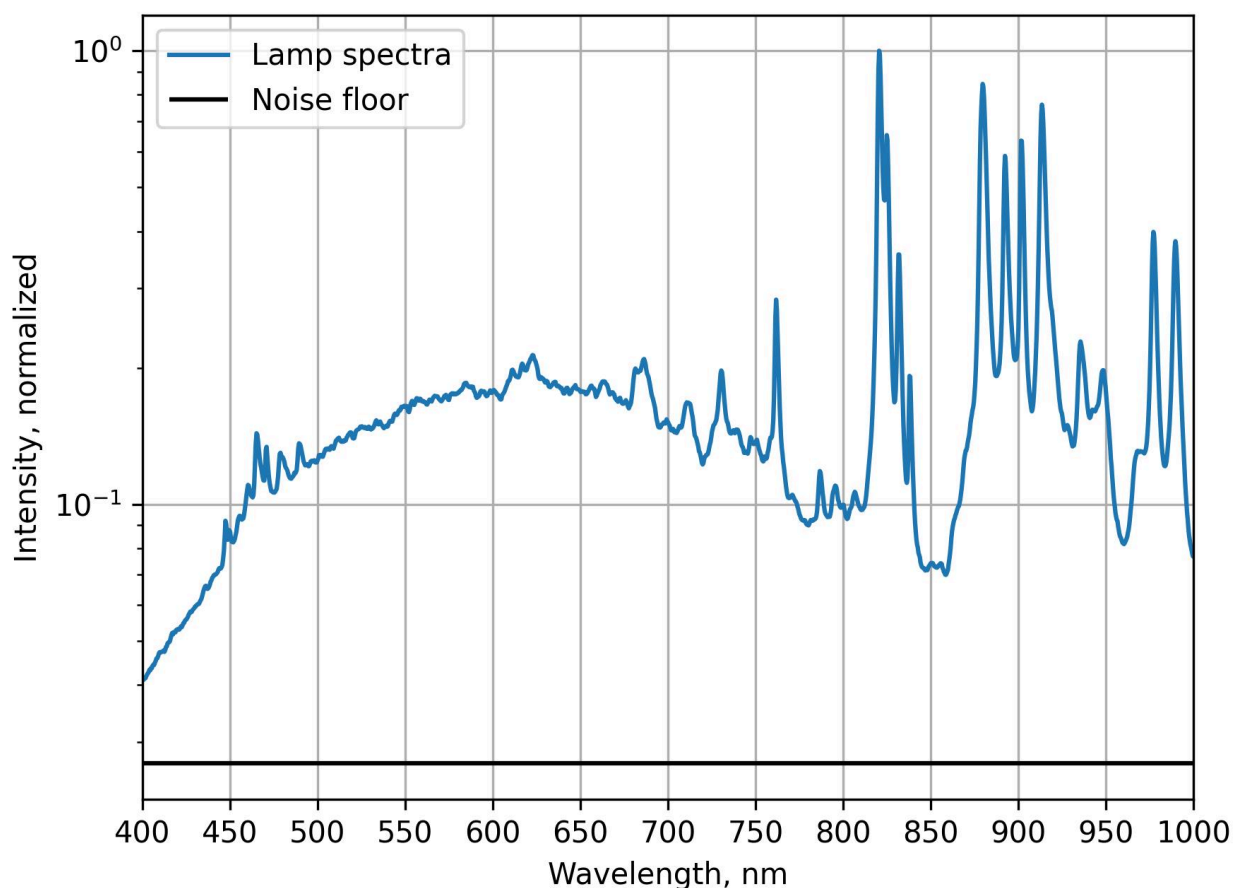


Figure 1. Normalized steady-state lamp spectrum (blue) versus spectrometer noise floor (black), 400–1000 nm, log intensity scale.

The spectrum is dominated by the broad Xenon continuum, several nanometers above the noise floor across the full visible-to-NIR window, with the characteristic Xe emission lines superimposed near 800–1000 nm.

Long-Term Measurement

To characterize lamp warm-up and long-term drift, we recorded one spectrum per second for 4 hours starting at ignition, averaging 10 scans per recorded sample at a fixed integration time (set once by autoexposure before the run).

Parameter	Value
Elapsed time	14400 s (4 hours)
Time step per measurement	1.0 s
Integration time	10.23 ms
Scans averaged per sample	10

Parameter	Value
Spectra array shape	(14400, 2048)

Warm-up and integrated power

For each time step, the spectrum was integrated over wavelength (trapezoidal rule) to obtain a single broadband power value $P(t)$.

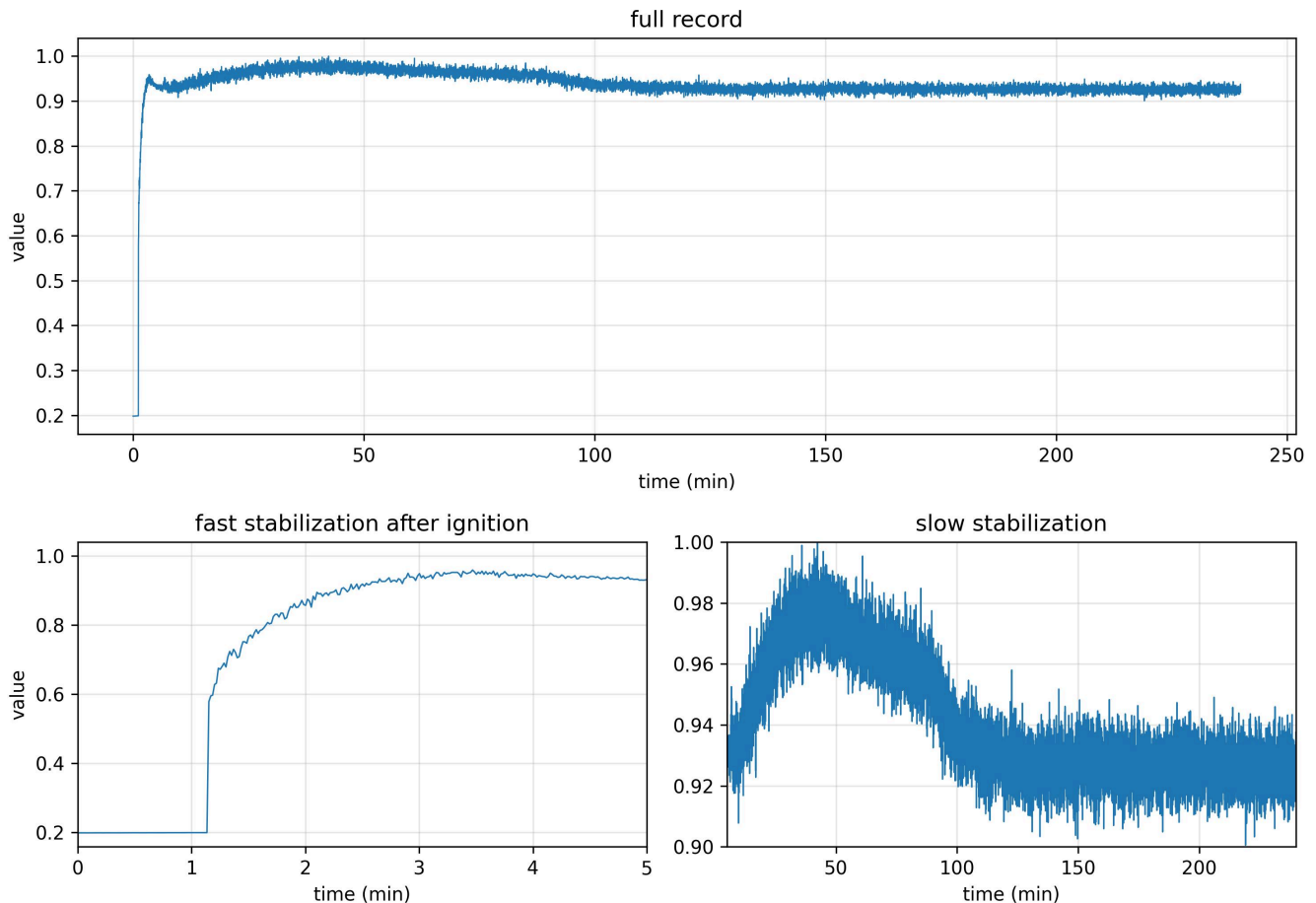


Figure 2. Integrated lamp power, normalized to its maximum, over the full 4-hour run. Top: full record. Bottom-left: ignition zoom (0–5 min) showing the fast ignition transient. Bottom-right: zoom of the slower warm-up and its settling to steady state (5 min to end of run, note the restricted y-range).

The trace shows a fast ignition transient over roughly the first 2 minutes, in which the lamp reaches ~90% of its eventual level. This is followed by a slower thermal **overshoot**, peaking around 40–50 minutes after ignition at a level a few percent above the eventual steady state, and then a gradual decay as the lamp continues to warm mechanically. The output does not settle onto a flat plateau until roughly **110–120 minutes** after ignition (because the plotted power is normalized to its maximum, which occurs during the overshoot, the steady-state plateau sits visibly below 1.0 in Figure 2). The

lamp should therefore not be treated as a stable reference source until ~2 hours after ignition, not simply after the initial transient has decayed.

Steady-state stability

Restricting to the settled tail of the run ($t \geq 7200$ s, i.e. the last 2 hours) isolates steady-state fluctuations from the warm-up transient.

Quantity	Value
Mean integrated power, $\bar{P}$	4.271e+06 (counts·nm)
Fractional RMS, $\sigma_P / \bar{P}$	6.984e-03
Peak-to-peak fractional deviation	6.204e-02

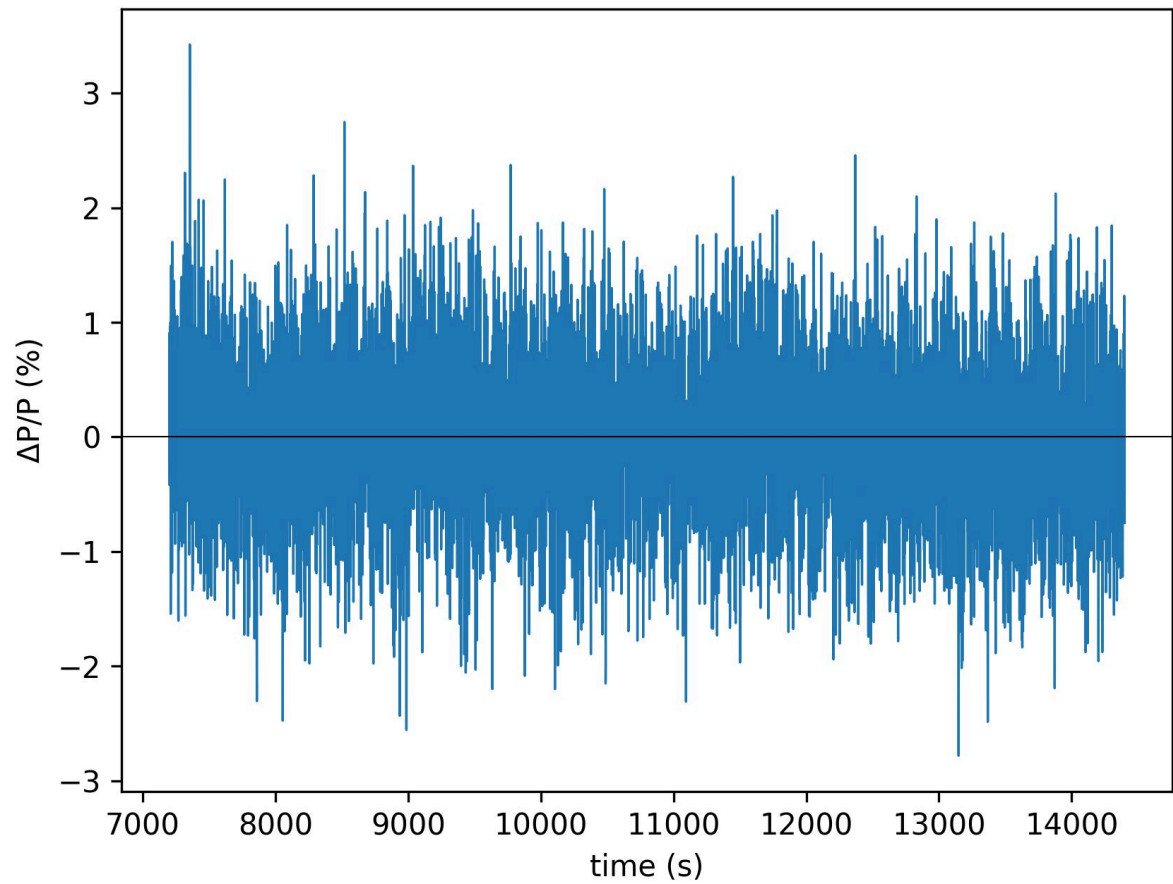


Figure 3. Fractional deviation of integrated power, $\Delta P/P$ (%), about its steady-state mean, for $t \geq 7200$ s.

Once warmed up, the lamp's broadband output is stable to within ~0.7% RMS, with occasional excursions up to ~6% peak-to-peak — consistent with slow

residual drift rather than random shot-to-shot noise.

Allan deviation

The overlapping Allan deviation, $\sigma_A(\tau)$, quantifies how the stability of the integrated power depends on the averaging (or reference) time τ used when comparing two measurements. It is computed by taking second differences of the cumulative sum of the fractional power deviation at spacing m samples, for a range of m , and forms the standard tool for separating noise sources that average down (white noise, whose $\sigma_A(\tau)$ falls as $\tau^{-1/2}$) from slower drifts that eventually dominate and cause $\sigma_A(\tau)$ to rise again at large τ . The minimum of the curve therefore identifies the optimal averaging/reference time: the point beyond which further averaging no longer improves stability because drift has taken over from noise.

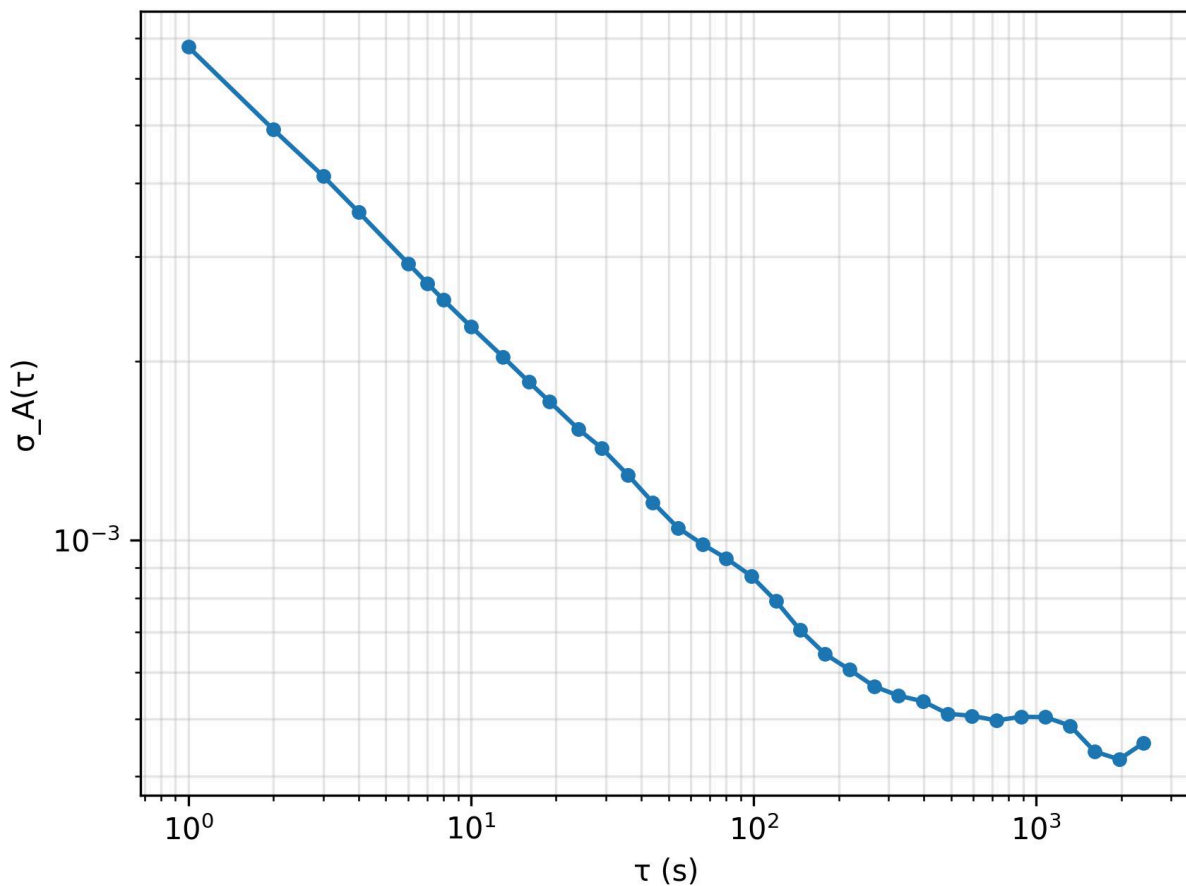


Figure 4. Overlapping Allan deviation of the fractional integrated power in steady state, versus averaging time τ .

The curve falls off as white noise ($\sigma_A(\tau) \propto \tau^{-1/2}$) from $\tau = 1$ s out to several hundred seconds, then its slope gradually shallows and it

saturates toward the end of the 2-hour steady-state window, reaching its lowest value, $\sigma_A = 4.275e-04$ (~0.04%), at $\tau = 1965$ s — comparable to the length of the segment itself. There is no true drift-driven upturn within this record: the slight uptick at the very last point reflects the small number of independent estimates available at that τ (an edge/statistics effect), not the lamp's power actually becoming less stable at long averaging times. In other words, over the 2-hour steady-state window probed here, slow drift never dominates over noise — averaging or re-referencing keeps paying off essentially all the way out to the length of the run.

Reference timing: how fast must the second measurement follow the first?

This curve is also the direct input for the timing budget of a sample/reference measurement pair. When a sample spectrum $C^{(s)}$ and a reference spectrum $C^{(r)}$ are acquired separated by a gap Δt and then divided, the ratio's error is set by how much the lamp happens to drift between the two acquisitions. Because the two acquisitions are effectively independent draws from the same underlying fluctuation, the uncorrelated error introduced into the ratio is

$$\sigma_{\text{pair}}(\Delta t) \approx \sqrt{2} \cdot \sigma_A(\Delta t)$$

i.e. the Allan deviation evaluated at $\tau = \Delta t$, scaled by $\sqrt{2}$ for the two independent measurements being differenced. This turns Figure 4 into a lookup table: pick the ratio-error budget you can tolerate, solve for the required $\sigma_A(\Delta t) < \text{budget} / \sqrt{2}$, and read the corresponding Δt off the curve — that is the maximum tolerable sample↔reference gap.

Concretely, for this lamp: a 0.2% budget on the ratio requires $\sigma_A(\Delta t) \leq 1.4e-3$, which the curve reaches within the first few tens of seconds; tightening the budget to 0.1% requires $\sigma_A(\Delta t) \leq 7e-4$, pushing the allowed gap out to order 100–200 s. The best achievable pair precision, using the curve's floor of $4.275e-4$ near $\tau \approx 1965$ s, is $\sqrt{2} \times 4.275e-4 \approx 6e-4$ (~0.06%) — but in practice a Δt of tens of seconds is the more useful operating point, since it already meets a sub-0.2% budget without forcing an impractically long gap between sample and reference acquisitions. The tighter the error budget, the sooner the reference must be re-acquired relative to the sample.

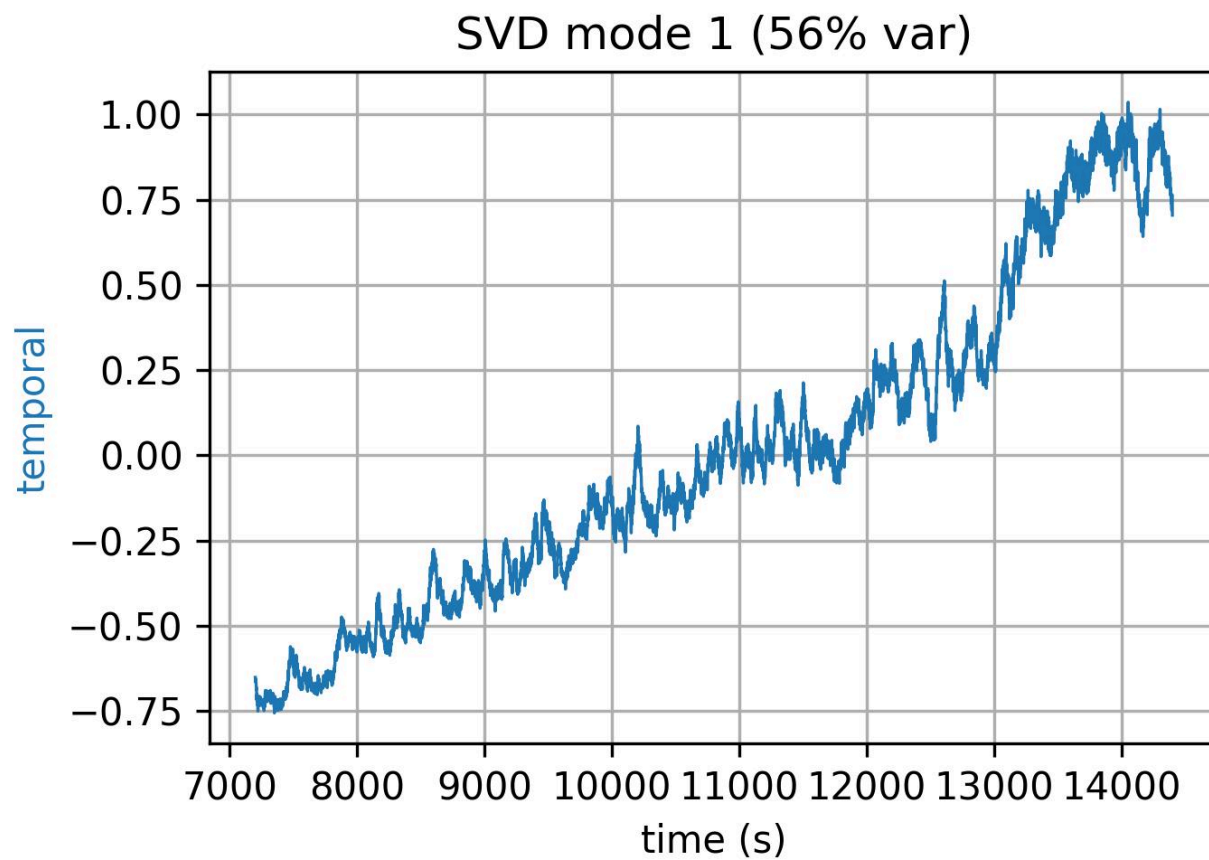
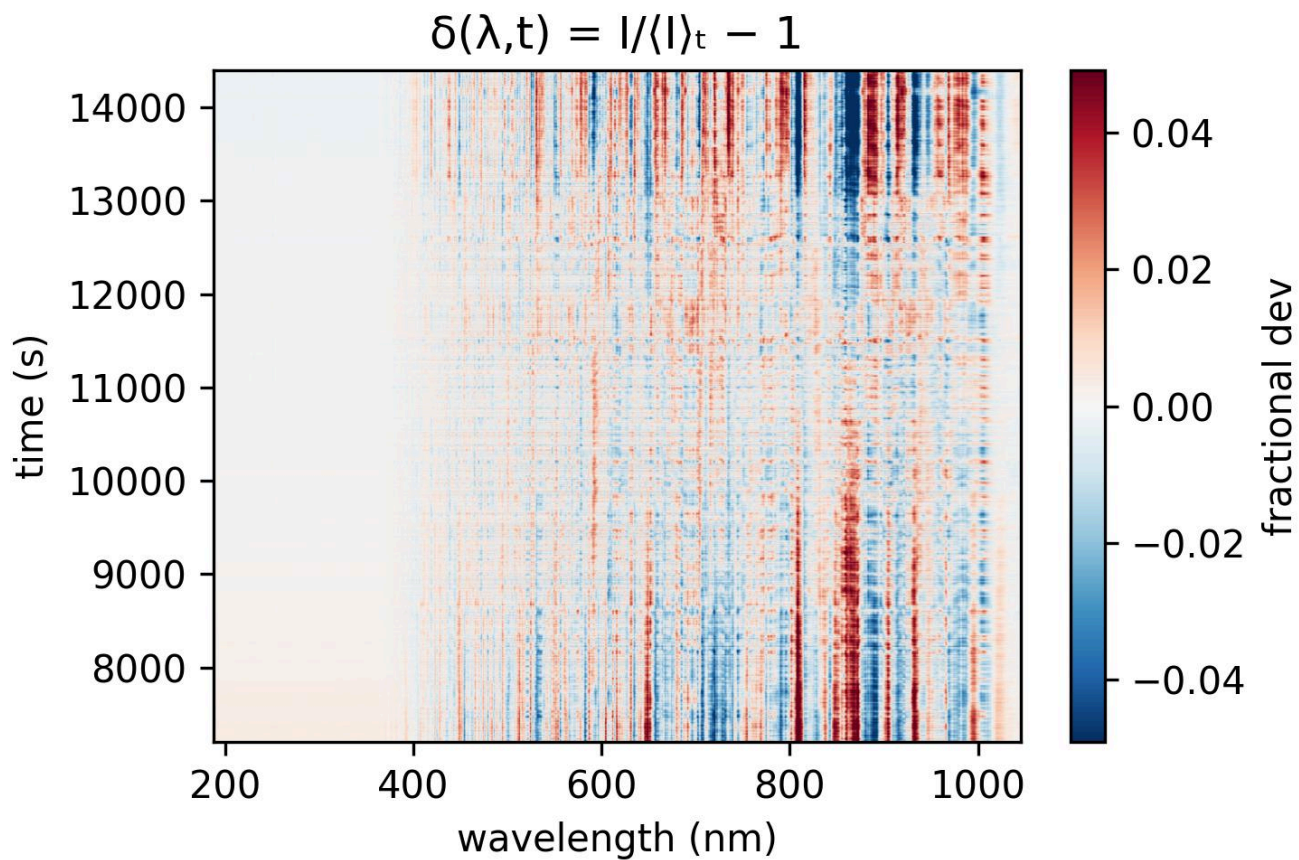
Amplitude-vs-shape decomposition

Steady-state drift can change the lamp output in two qualitatively different ways: a uniform brightness change across all wavelengths ("amplitude" drift), or a change in the relative shape of the spectrum ("shape" drift, e.g. color temperature shift). To separate these, we compute the fractional-deviation matrix

$$\delta(\lambda, t) = I(\lambda, t) / \langle I(\lambda) \rangle_t - 1$$

decompose it into a broadband amplitude term $a(t) = \langle \delta(\lambda, t) \rangle_\lambda$ (the mean deviation at each time, common to all wavelengths) and a shape residual $R(\lambda, t) = \delta(\lambda, t) - a(t)$, and compare their variances. As a cross-check, we also take the SVD of $\delta(\lambda, t)$: a dominant mode with near-zero spectral "flatness" (i.e. roughly constant across λ) indicates amplitude-like drift, while a mode with high spectral flatness (strong wavelength dependence) indicates shape-like drift.

Quantity	Value
Amplitude variance	3.164e-05 (16.5% of total)
Shape variance	1.606e-04 (83.5% of total)
SVD mode-1 variance fraction	0.56
SVD mode-1 spectral flatness	51.30



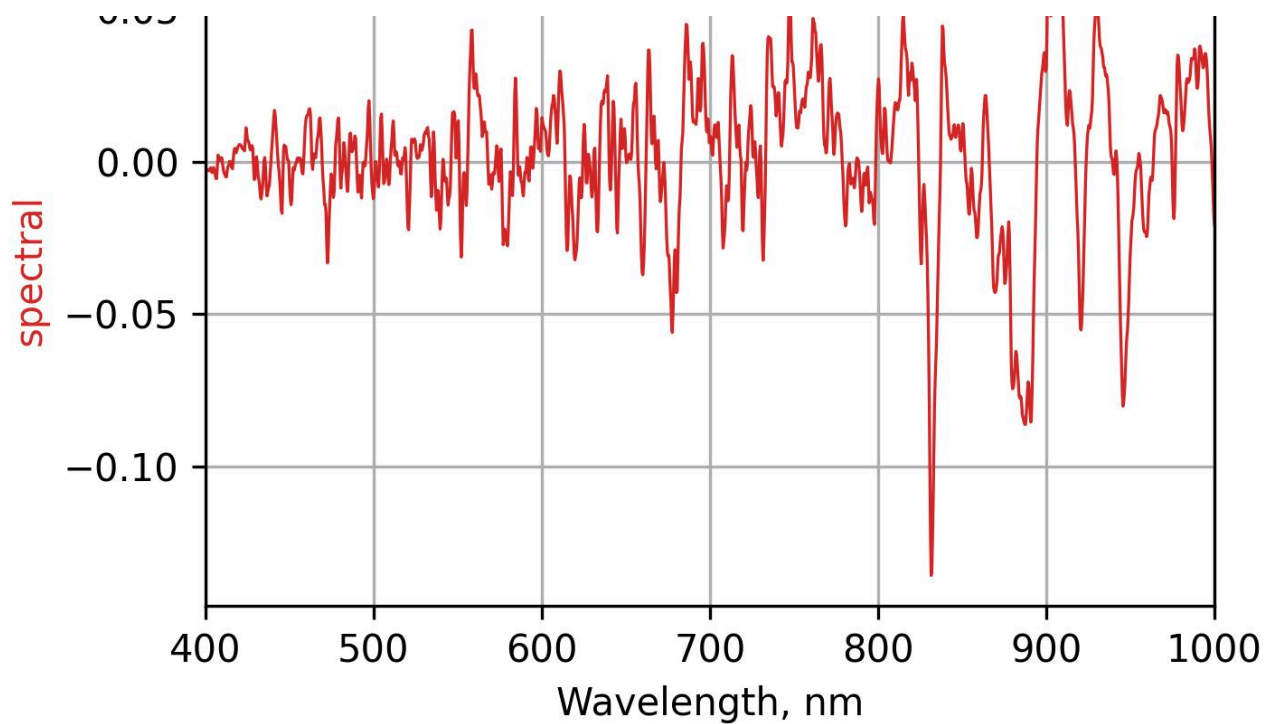


Figure 5. Top: fractional-deviation heatmap $\delta(\lambda, t)$. Middle: temporal profile of the dominant SVD mode. Bottom: spectral profile of the dominant SVD mode.

Steady-state drift is dominated by **spectral-shape** changes (83.5% of the variance) rather than uniform brightness changes (amplitude, 16.5%) — the opposite of a simple overall dimming or brightening. The dominant SVD mode alone accounts for 56% of the total variance and has a high spectral flatness (51.30, far from the ≈ 0 expected for a pure amplitude mode), confirming it is strongly wavelength-dependent. Its temporal profile (Figure 5, middle) is a slow, mostly monotonic ramp across the 2-hour tail rather than stationary noise, and its spectral profile (Figure 5, bottom) is largest in the red and NIR and peaks sharply near the Xe emission lines (800–1000 nm). Physically this points to a slow color-temperature drift (e.g. residual thermal drift of the lamp/monochromator after the ~2-hour warm-up) rather than a simple change in total output — consistent with the per-band noise pattern below.

Per-band noise

To see how the temporal noise found above is distributed across the spectrum, the steady-state spectra were projected onto a bank of Gaussian passbands (5 nm FWHM, spaced every 5 nm from 400–1000 nm), and the temporal coefficient of variation (CV, i.e. fractional std. dev. over the 2-hour tail) was computed for each band.

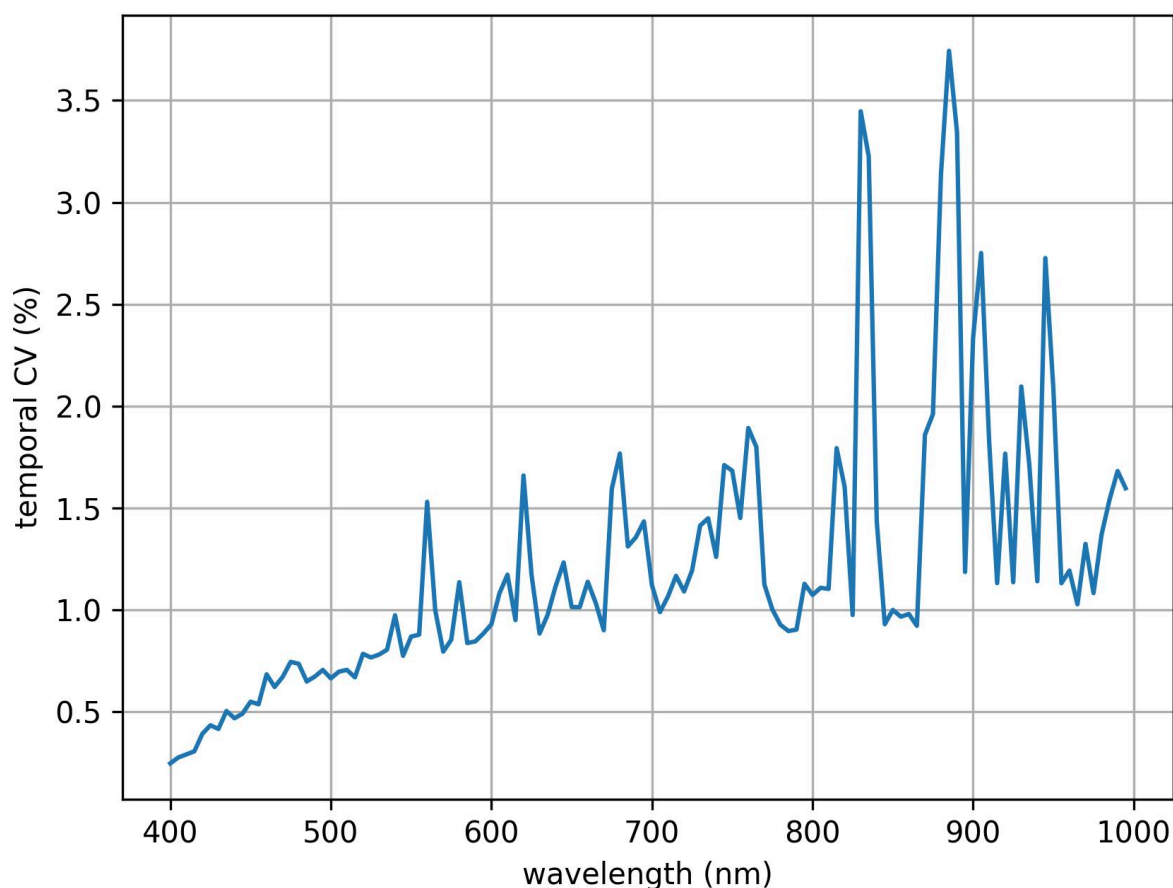


Figure 6. Temporal coefficient of variation, per 5-nm band, versus band center wavelength, over the steady-state tail ($t \geq 7200$ s).

The temporal CV rises steadily from ~0.25% in the blue (400 nm) to ~1–2% in the red (600–800 nm), with sharp spikes reaching 2–3.7% at the Xe emission lines (800–1000 nm). This is consistent with the amplitude/shape decomposition above: the blue continuum, where the dominant (shape-like) drift mode has the least spectral weight, is the most stable region of the spectrum, while the red/NIR continuum and the discrete Xe lines — where that mode has the most weight — carry most of the drift and noise.

Fast-Scale Measurement

In addition to the long-term run above, we acquired a 5-minute, unaveraged run at the spectrometer's maximum sampling rate to resolve fluctuations that the 10-scan averaging of the long-term run would smooth out. The monochromator (VIS grating, lamp baseline) and integration time (set once by autoexposure targeting 0.5 fill fraction before the run) were otherwise unchanged.

Parameter	Value
Elapsed time	300 s (5 min)
Integration time	7.2 ms
Scans averaged per sample	1 (none)
Time step per measurement	14.85 ms (~67 Hz)
Spectra array shape	(20207, 2048)

Source data:
[lamp_stability_spectral_fast_20260707_154730.npz](#) .

Quantity	Value
Mean integrated power, $\bar{P}$	3.419e+06 (counts·nm)
Fractional RMS, $\sigma_P / \bar{P}$	7.875e-03
Peak-to-peak fractional deviation	6.015e-02

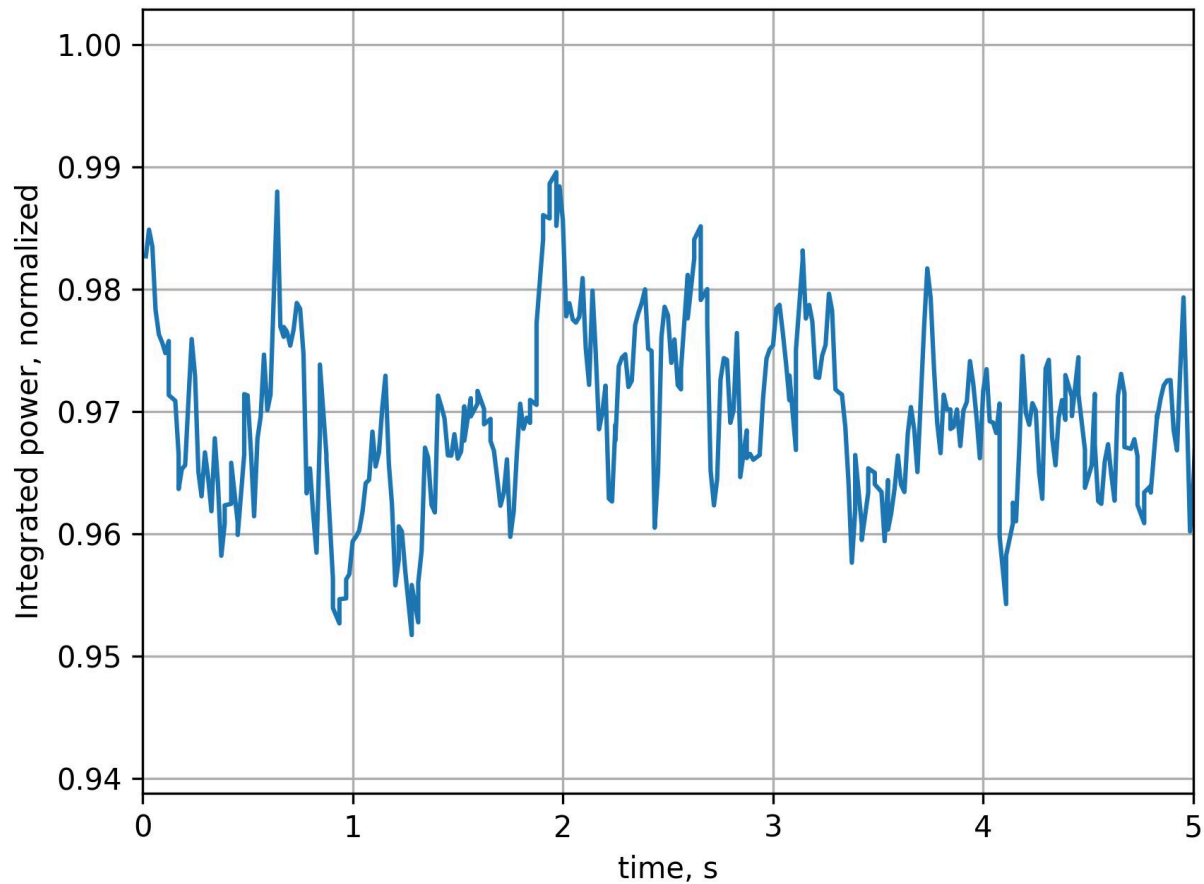


Figure 7. Integrated power, normalized to its maximum, over a 5-second window of the fast run.

At sub-second cadence the lamp shows $\sim 0.8\%$ RMS flicker (comparable to the $\sim 0.7\%$ RMS seen in the long-term steady-state run), confirming that most of the broadband fluctuation budget is already present on millisecond timescales rather than being an artifact of slower drift.

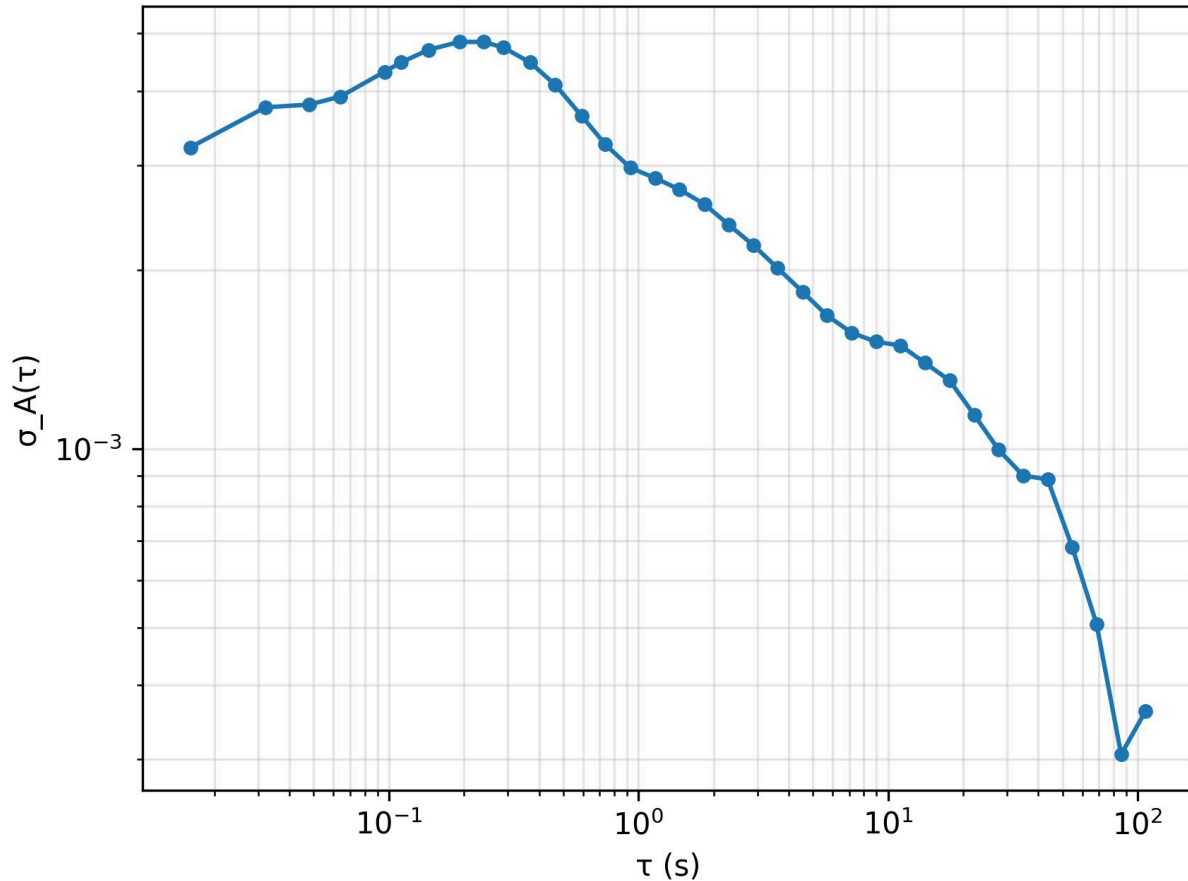


Figure 8. Overlapping Allan deviation of the fractional integrated power for the fast run, versus averaging time τ .

The Allan deviation shows a small correlated-flicker bump below $\tau \approx 0.2$ s, then falls off as white noise ($\sigma_A(\tau) \propto \tau^{-1/2}$) down to a minimum of $\sigma_A = 3.057e-04$ at $\tau = 86$ s, with only a slight uptick at the final, statistics-limited point. Unlike the long-term run, this 5-minute record is too short to probe whether a drift floor exists beyond $\tau \approx 100$ s; within the record, sub-second flicker averages down cleanly with no sign of a fixed noise plateau.

Results

- **Warm-up:** the lamp needs approximately **2 hours** after ignition to reach a stable steady-state output, and should not be used as a reference before then. Warm-up is non-monotonic: a fast rise over the first ~2 minutes is followed by a thermal overshoot peaking around 40–50 minutes, which then decays to a flat plateau only by ~110–120 minutes.
- **Steady-state stability:** once warmed up, integrated power is stable to ~0.7% RMS (up to ~6% peak-to-peak), and this fluctuation is dominated by

fast (sub-second) flicker rather than slow drift.

- **Allan deviation:** shows no drift-driven upturn within the 2-hour steady-state window — $\sigma_A(\tau)$ falls as white noise and only saturates near the end of the record, reaching $\sigma_A \approx 4.3 \times 10^{-4}$ (~0.04%) at $\tau \approx 1965$ s. Averaging or re-referencing therefore keeps helping across essentially the whole window.
- **Reference timing:** for a sample/reference power ratio, the pair error is $\sigma_{\text{pair}}(\Delta t) \approx \sqrt{2} \cdot \sigma_A(\Delta t)$; reading this off the Allan curve, a Δt of tens of seconds keeps the ratio error below ~0.2%, while ~100–200 s is needed to reach ~0.1%.
- **Amplitude vs. shape:** steady-state drift is dominated by **spectral-shape** changes (83.5% of variance) rather than uniform brightness changes (16.5%); the dominant drift mode is strongly wavelength-dependent (spectral flatness 51.3), consistent with a slow color-temperature drift concentrated in the red/NIR and at the Xe emission lines.
- **Per-band noise:** temporal noise is smallest in the blue (~0.25% CV at 400 nm) and rises to ~1–2% in the red, with 2–3.7% spikes at the Xe emission lines (800–1000 nm).
- **Fast-timescale stability:** sub-second flicker is ~0.8% RMS and averages down as white noise to $\sigma_A \approx 3.1 \times 10^{-4}$ by $\tau \approx 86$ s, with no sign of a fast-noise floor within the 5-minute fast run.